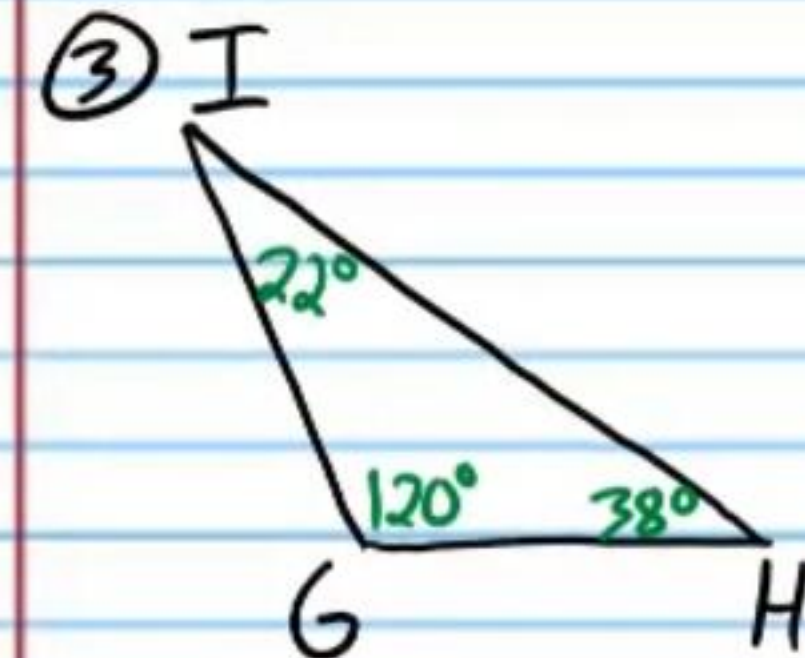
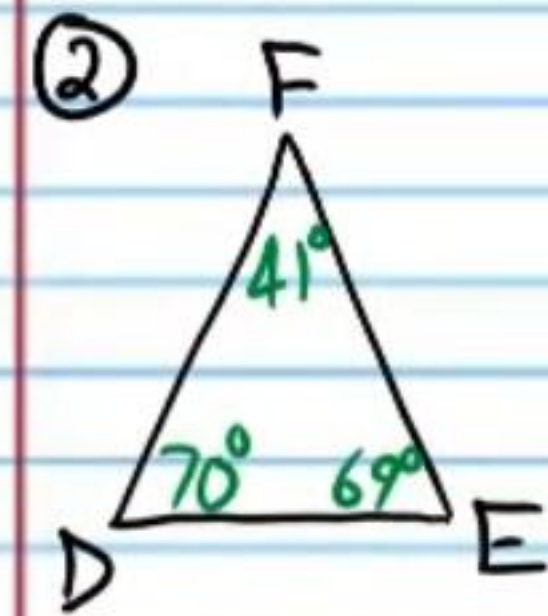
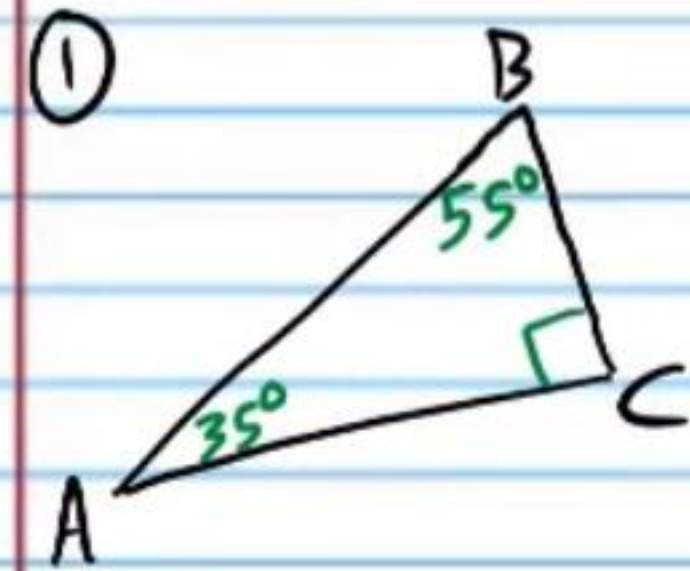


Ratios, Proportions, and Triangle Inequality Theorems (Homework)

List the sides of the triangles from shortest to longest.



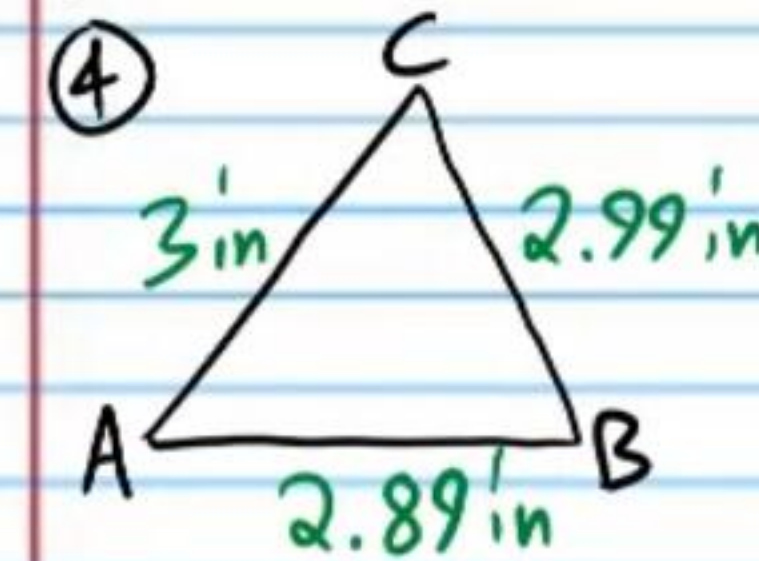
Answers

BC, AC, AB

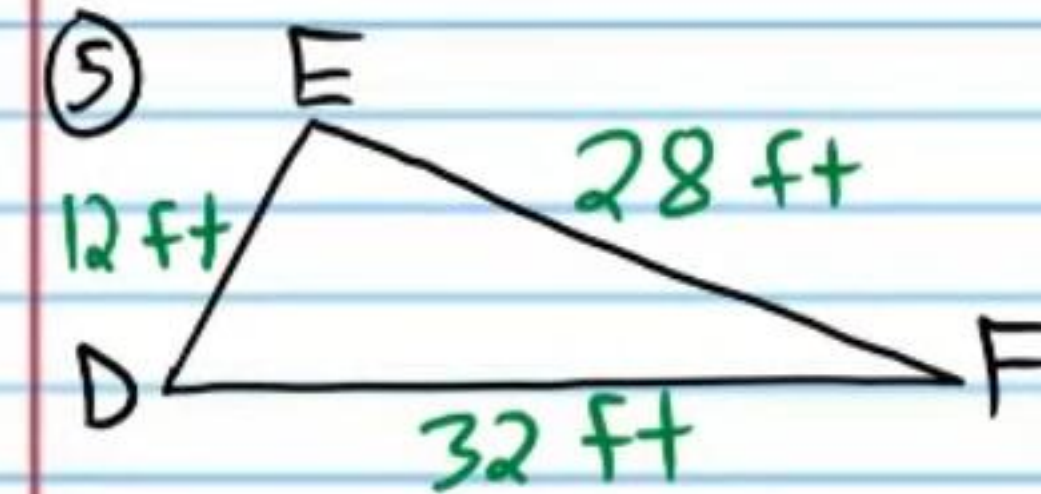
DE, DF, FE

GH, GI, HI

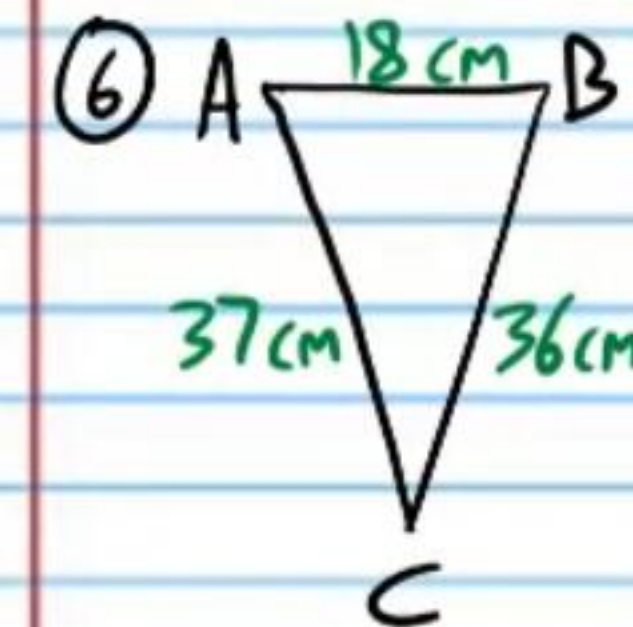
List the angles of the triangles from least to greatest.



$\angle C, \angle A, \angle B$



$\angle F, \angle D, \angle E$



$\angle C, \angle A, \angle B$

Is it possible to make a triangle with each group of side lengths? If not, write the inequality that shows that it's not possible.

⑦ 5, 10, 7

Yes

⑧ 1, 6, 3

No, $1+3 \nless 6$

⑨ 4, 5, 16

No, $4+5 \nless 16$

⑩ 11, 8, 7

Yes

⑪ 6, 3, 3

No, $3+3 \nless 6$

If each pair of numbers below represents side lengths of a triangle, write an inequality that shows the possible lengths of the third side x .

⑫ 7, 5

$2 < x < 12$

⑬ 100, 60

$40 < x < 160$

⑭ 3, 10

$7 < x < 13$

⑮ 21, 37

$16 < x < 58$

⑯ I have three pencils and five pens. Write ratios of these numbers in six different ways.

3:5, 5:3
3 to 5, 5 to 3
 $\frac{3}{5}$, $\frac{5}{3}$

⑰ Matt has \$15 and Larry has \$6. Write ratios of these numbers in lowest terms in six different ways.

5:2, 2:5
5 to 2, 2 to 5
 $\frac{5}{2}$, $\frac{2}{5}$

Reduce the following ratios to lowest terms.

⑱ 8 to 18.

4 to 9

⑲ 2:10

1:5

⑳ $4\frac{9}{7}$

$7\frac{1}{1}$

㉑ 17 to 17

1 to 1

Determine if the proportions are true.

㉒ $\frac{5}{7} \stackrel{?}{=} \frac{21}{35}$

No

㉓ $\frac{8}{25} \stackrel{?}{=} \frac{4}{50}$

No

㉔ $\frac{15}{40} \stackrel{?}{=} \frac{3}{8}$

Yes

㉕ $\frac{4}{12} \stackrel{?}{=} \frac{5}{15}$

Yes

㉖ $\frac{80}{20} \stackrel{?}{=} \frac{40}{10}$

Yes

Solve the proportions.

㉗ $\frac{7}{2} = \frac{a}{5}$

$a = 17.5$

$$\textcircled{28} \frac{3}{b} = \frac{9}{15}$$

$$b=5$$

$$\textcircled{29} \frac{c}{6} = \frac{5}{9}$$

$$c = 3.\bar{3} \text{ or } 3\frac{1}{3}$$

$$\textcircled{30} \frac{4}{7} = \frac{12}{d}$$

$$d=21$$

Solve the following problems by writing proportion statements (i.e. equations) and using them to find the answers.

$\textcircled{31}$ If a turtle crawls 6 miles in 11 hours, how far will it walk in 88 hours?

$$\frac{6}{11} = \frac{D}{88}, D=48 \text{ mi}$$

$\textcircled{32}$ If grapes cost \$8 for 3 pounds, how many pounds of grapes could you buy with \$20?

$$\frac{8}{3} = \frac{20}{P}, P=7.5 \text{ lbs}$$